Null Hypothesis (H_0) and Alternative Hypothesis (H_a)

1. Concept of Hypothesis

A **hypothesis** is an assumption or claim about a **population parameter** (such as the mean or proportion) that is tested using statistical methods.

There are **two types** of hypotheses:

1. **Null Hypothesis (H_0)** → Represents **no effect, no difference, or the status quo**. It assumes that any observed difference is due to chance.
2. **Alternative Hypothesis (H_a or H_1)** → Represents a **real effect, a difference, or change**. It is what we want to prove.

Key Differences

Feature	Null Hypothesis (H_0)	Alternative Hypothesis (H_a)
Definition	No change, effect, or difference	A real change, effect, or difference exists
Symbol	H_0	H_a or H_1
Example (Mean Test)	$H_0 : \mu = 50$ (No difference)	$H_a : \mu \neq 50$ (There is a difference)
Example (Proportion Test)	$H_0 : p = 0.6$ (No change)	$H_a : p \neq 0.6$ (Change in proportion)
Goal	We try to reject it (or fail to reject)	We try to prove it
Basis for Decision	If p-value $\leq \alpha$, reject H_0	If p-value $> \alpha$, fail to reject H_0

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2. Types of Alternative Hypothesis

1. **Two-Tailed Hypothesis**:
   - Tests for **any difference** (increase or decrease).
   - Example:
     - $H_0 : \mu = 50$
     - $H_a : \mu \neq 50$

2. **One-Tailed (Right-Tailed) Hypothesis**:
    - Tests if the parameter is **greater** than a certain value.
    - Example:
        - $H_0 : \mu \leq 50$
        - $H_a : \mu > 50$
3. **One-Tailed (Left-Tailed) Hypothesis**:
    - Tests if the parameter is **less** than a certain value.
    - Example:
        - $H_0 : \mu \geq 50$
        - $H_a : \mu < 50$

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3. Example Problems

Example 1: Mean Test (Two-Tailed)

Problem:
A company claims that its batteries last **300 hours** on average. A consumer group tests **40 batteries** and finds an average lifespan of **290 hours** with a standard deviation of **50 hours**. At $\alpha = 0.05$, test the claim.

Step 1: Define Hypotheses

- $H_0 : \mu = 300$ (No difference)
- $H_a : \mu \neq 300$ (Battery life is different)

Step 2: Compute Test Statistic (Z-Test)

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

$$Z = \frac{290 - 300}{50 / \sqrt{40}}$$

$$Z = \frac{-10}{7.91} = -1.26$$

Step 3: Compare with Critical Value

- **For $\alpha = 0.05$ (Two-Tailed), Z-Critical = ±1.96**
- Since **-1.26 is within (-1.96, 1.96)**, we **fail to reject H_0**.

Conclusion:

There is **not enough evidence** to say that battery life is different from 300 hours.

Example 2: Proportion Test (One-Tailed)

Problem:
A school claims that **60%** of students pass a math exam. A researcher believes it is **less** than 60%. In a sample of **200 students**, **108 passed**. Test the claim at $\alpha = 0.05$.

Step 1: Define Hypotheses

- $H_0 : p = 0.6$ (No difference)
- $H_a : p < 0.6$ (Fewer students pass)

Step 2: Compute Test Statistic (Z-Test for Proportion)

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

where:

- $\hat{p} = \frac{108}{200} = 0.54$
- $p = 0.6$
- $n = 200$

$$Z = \frac{0.54 - 0.6}{\sqrt{\frac{0.6(0.4)}{200}}}$$

$$Z = \frac{-0.06}{\sqrt{0.0012}} = \frac{-0.06}{0.0346} = -1.73$$

Step 3: Compare with Critical Value

- **For $\alpha = 0.05$ (One-Tailed, Left), Z-Critical = -1.645**
- Since **-1.73 < -1.645**, we **reject** H_0.

Conclusion:

There is **significant evidence** that the pass rate is **less than 60%**.

Example 3: One-Tailed Test for Weight Loss Drug

Problem:
A weight loss drug claims that users lose **at least 5 kg** after a month. A test on **30 patients** finds an average weight loss of **4.2 kg** with a standard deviation of **1.5 kg**. Test the claim at $\alpha = 0.01$.

Step 1: Define Hypotheses

- $H_0 : \mu \geq 5$ (The drug works as claimed)
- $H_a : \mu < 5$ (The drug is ineffective)

Step 2: Compute Test Statistic (t-Test)

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

$$t = \frac{4.2 - 5}{1.5/\sqrt{30}}$$

$$t = \frac{-0.8}{0.2739} = -2.92$$

Step 3: Compare with Critical Value

- **For** $\alpha = 0.01$, $df = 29$, **t-Critical = -2.462**
- Since **-2.92 < -2.462**, we **reject** H_0.

Conclusion:

There is **strong evidence** that the drug does **not help patients lose 5 kg** as claimed.

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4. Summary Table

Test Type	Null Hypothesis (H_0)	Alternative Hypothesis (H_a)	Example
Two-Tailed	$H_0 : \mu = \mu_0$	$H_a : \mu \neq \mu_0$	Checking if battery life differs from 300 hours.
Left-Tailed	$H_0 : \mu \geq \mu_0$	$H_a : \mu < \mu_0$	Testing if a drug reduces weight by at least 5 kg.
Right-Tailed	$H_0 : \mu \leq \mu_0$	$H_a : \mu > \mu_0$	Checking if a new teaching method improves scores.